

Panel IV Results – Specification 1: Wage only

Treatment: real AEWR (2012 dollars). County and year fixed effects; standard errors clustered by AEWR–region x target subregion.

Donor wage level: county–mapped OEWS–area Big–Six hourly–wage proxy; QCEW supplies county employment and calibration features; source year t–1.

	H–2A outcomes						Farm outcomes					
Statistic	Certified workers / 2011 farm emp.	Certified hours / 2011 farm emp.	Applications / 2011 farm employment	Employers / 2011 farm employment	Certified positions / application	Certified hours / position	Real crop price index	Farm employment	Production expense share	Real farm income / worker	Hired–labor expense share	Crop output quantity index
AEWR coefficient	–0.0963*	–57.57*	–0.0014	–0.0026	0.8100	–116.9**	–4.212	–8.671	–0.0261	36,016**	0.0063**	–11.09
Clustered standard error	(0.0536)	(34.29)	(9.19e–04)	(0.0034)	(3.372)	(53.06)	(6.573)	(10.54)	(0.0302)	(14,107)	(0.0030)	(134.8)
p–value	0.077	0.097	0.124	0.447	0.811	0.030	0.523	0.413	0.390	0.013	0.041	0.935
First–stage excluded–instrument F	23.02	23.02	23.02	23.02	19.42	19.31	22.97	23.02	23.02	23.02	23.02	22.97
Observations	33,497	33,497	33,497	33,497	21,118	20,825	33,441	33,497	33,497	33,497	33,497	33,441
Counties	2,848	2,848	2,848	2,848	2,553	2,538	2,847	2,848	2,848	2,848	2,848	2,847
Inference clusters	80	80	80	80	80	80	80	80	80	80	80	80

Notes: Coefficients report the effect of a one–dollar increase in real AEWR. Clustered standard errors are in parentheses; * p<0.10, ** p<0.05, *** p<0.01. The first–stage F is the squared clustered t statistic for the excluded instrument.

Panel IV Results – Specification 2: Seasonal/composition

Treatment: real AEWR (2012 dollars). County and year fixed effects; standard errors clustered by AEWR–region x target subregion.

Donor wage level: county–mapped OEWS–area Big–Six hourly–wage proxy; QCEW supplies county employment and calibration features; source year t–1.

	H–2A outcomes						Farm outcomes					
Statistic	Certified workers / 2011 farm emp.	Certified hours / 2011 farm emp.	Applications / 2011 farm employment	Employers / 2011 farm employment	Certified positions / application	Certified hours / position	Real crop price index	Farm employment	Production expense share	Real farm income / worker	Hired–labor expense share	Crop output quantity index
AEWR coefficient	–0.0849*	–54.36*	–0.0022	–0.0037	0.7817	–88.17*	–2.626	–13.08	–7.57e–04	36,066***	0.0055*	1.667
Clustered standard error	(0.0477)	(29.83)	(0.0013)	(0.0038)	(2.883)	(47.15)	(5.925)	(10.32)	(0.0296)	(12,881)	(0.0028)	(120.3)
p–value	0.079	0.072	0.109	0.335	0.787	0.065	0.659	0.209	0.980	0.006	0.051	0.989
First–stage excluded–instrument F	39.42	39.42	39.42	39.42	30.90	30.83	39.31	39.42	39.42	39.42	39.42	39.31
Observations	33,497	33,497	33,497	33,497	21,118	20,825	33,441	33,497	33,497	33,497	33,497	33,441
Counties	2,848	2,848	2,848	2,848	2,553	2,538	2,847	2,848	2,848	2,848	2,848	2,847
Inference clusters	80	80	80	80	80	80	80	80	80	80	80	80

Notes: Coefficients report the effect of a one–dollar increase in real AEWR. Clustered standard errors are in parentheses; * p<0.10, ** p<0.05, *** p<0.01. The first–stage F is the squared clustered t statistic for the excluded instrument.

Panel IV Results – Specification 3: Wage only + controls

Treatment: real AEWR (2012 dollars). County and year fixed effects; standard errors clustered by AEWR–region x target subregion.

Donor wage level: county–mapped OEWS–area Big–Six hourly–wage proxy; QCEW supplies county employment and calibration features; source year t–1.

	H–2A outcomes						Farm outcomes					
Statistic	Certified workers / 2011 farm emp.	Certified hours / 2011 farm emp.	Applications / 2011 farm employment	Employers / 2011 farm employment	Certified positions / application	Certified hours / position	Real crop price index	Farm employment	Production expense share	Real farm income / worker	Hired–labor expense share	Crop output quantity index
AEWR coefficient	–0.1099*	–69.95*	–0.0011	–0.0015	–1.846	–114.9**	–4.397	–4.811	–0.0238	37,522**	0.0064*	10.10
Clustered standard error	(0.0563)	(36.72)	(9.00e–04)	(0.0031)	(3.425)	(57.22)	(6.710)	(9.785)	(0.0335)	(14,399)	(0.0032)	(175.6)
p–value	0.054	0.060	0.227	0.626	0.592	0.048	0.514	0.624	0.481	0.011	0.052	0.954
First–stage excluded–instrument F	24.91	24.91	24.91	24.91	20.33	20.05	24.96	24.91	24.91	24.91	24.91	24.96
Observations	33,497	33,497	33,497	33,497	21,118	20,825	33,441	33,497	33,497	33,497	33,497	33,441
Counties	2,848	2,848	2,848	2,848	2,553	2,538	2,847	2,848	2,848	2,848	2,848	2,847
Inference clusters	80	80	80	80	80	80	80	80	80	80	80	80

Notes: Coefficients report the effect of a one–dollar increase in real AEWR. Clustered standard errors are in parentheses; * p<0.10, ** p<0.05, *** p<0.01. The first–stage F is the squared clustered t statistic for the excluded instrument.

Panel IV Results – Specification 4: Seasonal/composition + controls (preferred)

Treatment: real AEWR (2012 dollars). County and year fixed effects; standard errors clustered by AEWR–region x target subregion.

Donor wage level: county–mapped OEWS–area Big–Six hourly–wage proxy; QCEW supplies county employment and calibration features; source year t–1.

	H–2A outcomes						Farm outcomes					
Statistic	Certified workers / 2011 farm emp.	Certified hours / 2011 farm emp.	Applications / 2011 farm employment	Employers / 2011 farm employment	Certified positions / application	Certified hours / position	Real crop price index	Farm employment	Production expense share	Real farm income / worker	Hired–labor expense share	Crop output quantity index
AEWR coefficient	–0.0988**	–67.07**	–0.0019	–0.0025	–1.961	–81.38	–3.077	–9.905	0.0018	38,511***	0.0054*	22.32
Clustered standard error	(0.0489)	(32.45)	(0.0013)	(0.0036)	(2.883)	(51.96)	(5.877)	(9.330)	(0.0315)	(13,022)	(0.0029)	(159.1)
p–value	0.047	0.042	0.162	0.500	0.498	0.121	0.602	0.292	0.954	0.004	0.064	0.889
First–stage excluded–instrument F	46.00	46.00	46.00	46.00	34.11	33.79	46.11	46.00	46.00	46.00	46.00	46.11
Observations	33,497	33,497	33,497	33,497	21,118	20,825	33,441	33,497	33,497	33,497	33,497	33,441
Counties	2,848	2,848	2,848	2,848	2,553	2,538	2,847	2,848	2,848	2,848	2,848	2,847
Inference clusters	80	80	80	80	80	80	80	80	80	80	80	80

Notes: Coefficients report the effect of a one–dollar increase in real AEWR. Clustered standard errors are in parentheses; * p<0.10, ** p<0.05, *** p<0.01. The first–stage F is the squared clustered t statistic for the excluded instrument.